Precision Coatings Inc.

Excellence in Surface Technology

WORK ORDER

To:	Pneumatic Controls Engineering Department
From:	Dr. Evelyn Reed, Head of Production
Date:	September 9, 2025
Project No.:	PC-CC-2025-05
Subject:	**Design of a Pneumatic Control System for an Automated Component Coating and Curing Station**

1. Project Overview

Precision Coatings Inc. is developing a new automated workstation to apply a specialized protective coating to high-value components. To ensure maximum consistency and eliminate manual handling, the entire process—from loading to coating to curing—must be automated. We have finalized the mechanical design and are now commissioning your department to develop the complete pneumatic control system that will orchestrate the machine's movements.

2. Scope of Work & Deliverables

Your team is tasked with the full design and documentation of the pneumatic control circuit. A final design portfolio must be submitted, containing:

1. **Sequence Deduction:** A formal declaration of the operational sequence (e.g., A+ B- C+...) derived from the requirements in Section 4.
2. **Circuit Analysis:** A brief report detailing the grouping of the sequence and the rationale for the design choices made.
3. **Bill of Materials (BOM):** A complete component list, including all actuators, valves, sensors, and accessories.
4. **Space-State Graph:** A displacement-step diagram for all actuators in the system.
5. **Pneumatic Circuit Schematic:** The complete circuit diagram, drawn to ISO 1219 standards.

3. System Components

The station utilizes **five double-acting pneumatic cylinders**, each with a specific function:

- **Cylinder A:** A **Vertical Clamp** that extends downwards to secure the component.
- **Cylinder B:** A **Coating Nozzle Slide** that moves the spray nozzle horizontally.
- **Cylinder C:** A **Component Flipper** that rotates the component 180 degrees.
- **Cylinder D:** A **Horizontal Transfer Slide** that moves the component into the workstation.
- **Cylinder E:** A **Vertical Curing Lift** that moves the component up into a curing chamber.

4. Operational Requirements

The system must execute the following five stages in a single, continuous cycle:

Stage 1: Loading and First Coat

a. First, the **transfer slide (D) extends** to bring a raw component into the coating bay.
b. Next, the **vertical clamp (A) extends** to hold the component firmly in place.
c. Finally, the **coating nozzle (B) extends**, applying the first layer of coating.

Stage 2: Repositioning

a. The **vertical clamp (A) retracts** just enough to release its grip.
b. The **transfer slide (D) retracts** to clear the workspace.
c. The **component flipper (C) extends**, turning the component over.

Stage 3: Second Coat

a. The **vertical clamp (A) extends** once more to re-secure the flipped component.
b. The **coating nozzle (B) retracts** to its home position, applying the second coat.

Stage 4: Transfer to Curing

a. The **vertical clamp (A) retracts**, fully releasing the coated component.
b. The **component flipper (C) retracts** to its home position.
c. The **curing lift (E) extends** upwards into the curing chamber.

Stage 5: Curing and Reset

a. The **curing lift (E) retracts** to its home position, which completes the pneumatic cycle. The component remains in the chamber for a duration controlled by a separate timer.

5. Design Constraints & Safety Mandates

- **Control Methodology:** Due to the highly sequential nature of this process, the control circuit **must be designed using the Step-by-Step Method**.
- **Operator Safety:** For maintenance and setup, the machine cycle must be initiated by the **simultaneous activation of two pushbuttons**.
- **External Signal:** The timer for the curing process is an external system. The signal from the limit switch that confirms the **curing lift (E) is extended** will be used to start this external timer.

Please submit the complete design portfolio for review by **September 12, 2025**.

Sincerely,

Dr. Evelyn Reed

Head of Production
Precision Coatings Inc.